EE6094
CAD for VLSI Design

Bonus Exercise

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Slides Credit: TA Yin-Rong Zhuo

- version1 20250227
- version2 20250305

Introduction

- This exercise provides an opportunity for you to use commercial EDA tools from Synopsys for front-end design procedure
  - Design Compiler
- Use the Verilog file you generated to run logic synthesis
  - With Planner technology
  - With FinFET technology
- You need to answer all questions during the bonus exercise
- The bonus will be at most 10%

Requirement

- You are asked to submit a **report** for your bonus exercise assignment
  - Separate file
- The report should at least include:
  - The Answer of each questions
  - The difficulties you encountered and how you solve the problems
  - What you learn from this assignment
  - Suggestions of this bonus assignment
- Name your file as StudID_Name_bonus_report.pdf (ex: 9862534_陳聿廣_bonus_report.pdf).
  - Note that the only acceptable report file format is .pdf, no .doc/.docx or other files are acceptable.
  - Please upload your report to ee-class before deadline

Requirement

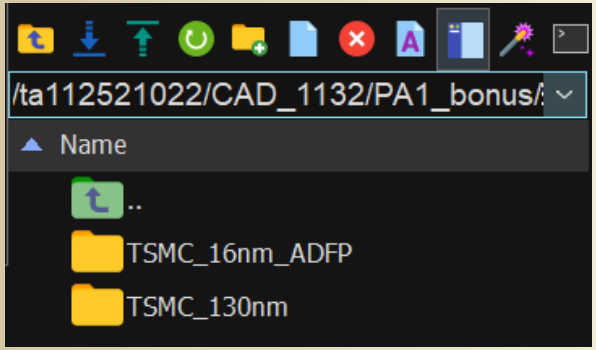
- Also, submit the **output files** onto eeclass
  - Verilog file after synthesis
  - Standard Delay File (SDF)
  - Area report
  - Power report

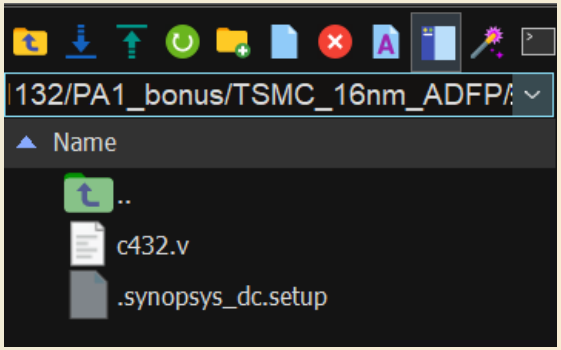
Requirement

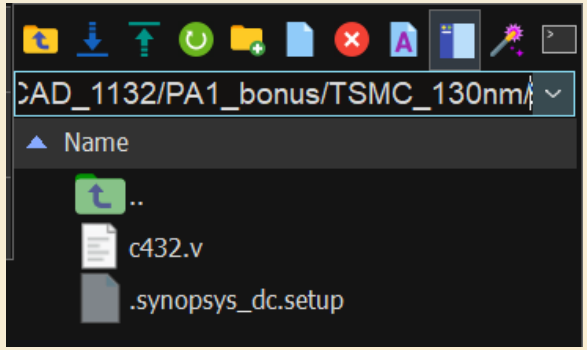
- Please put all required output files in a folder, the name of the folder is PA1_bonus_StudentID
- Use the below command to compress the folder in the linux environment, and the compressed file name should be the same as PA1_bonus_StudentID
  - tar cvf PA1_bonus_StudentID.tar PA1_bonus_StudentID

Download File

1. Download file “PA1_bonus.tar.gz” from ee-class.
2. Upload “PA1_bonus.tar.gz” to server.
3. Unzip file “PA1_bonus.tar.gz”, $ tar zxvf PA1_bonus.tar.gz
4. $ cd PA1_bonus
5. Copy your Verilog files (c432.v) into these two folders.







6. Do the following steps two times to compare both results.
7. Change into one of the directory. (ex. $ cd TSMC_16nm_ADFP)

Design Compiler
(Front End Design Procedure)

Design Compiler(DC)

1. Source license
   $ source /usr/cad/synopsys/CIC/synthesis.cshrc
2. Active Design Compiler
   $ dv&

Design Compiler(DC)

DC command line

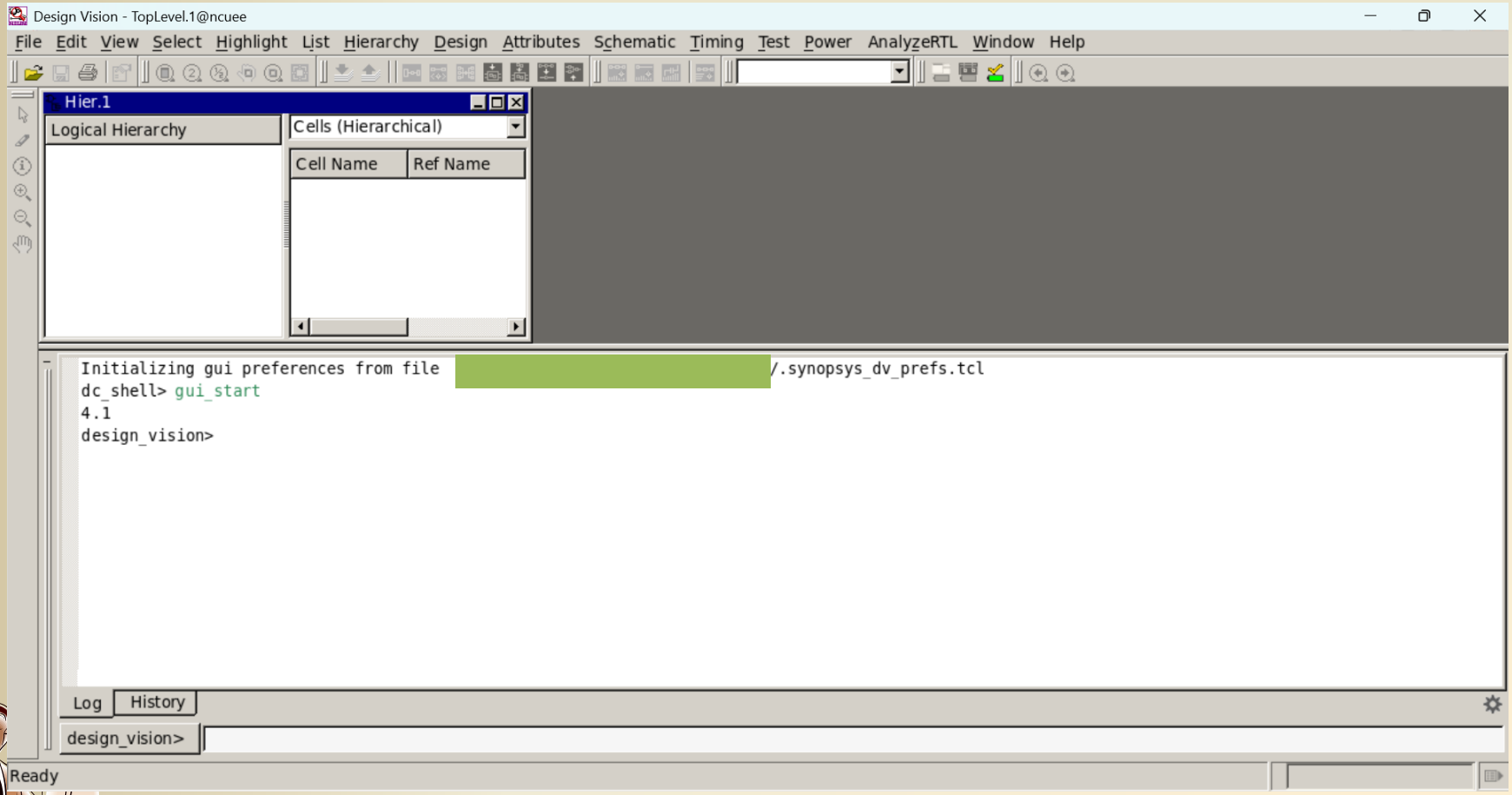
```
                    Design Compiler Graphical
                          DC Ultra (TM)
                           DFTMAX (TM)
                        Power Compiler (TM)
                          DesignWare (R)
                          DC Expert (TM)
                         Design Vision (TM)
                         HDL Compiler (TM)
                         VHDL Compiler (TM)
                            DFT Compiler
                         Design Compiler(R)

            Version P-2019.03-SP1-1 for linux64 - May 06, 2019

                  Copyright (c) 1988 - 2019 Synopsys, Inc.
   This software and the associated documentation are proprietary to Synopsys,
 Inc. This software may only be used in accordance with the terms and conditions
 of a written license agreement with Synopsys, Inc. All other use, reproduction,
            or distribution of this software is strictly prohibited.
Initializing...
```

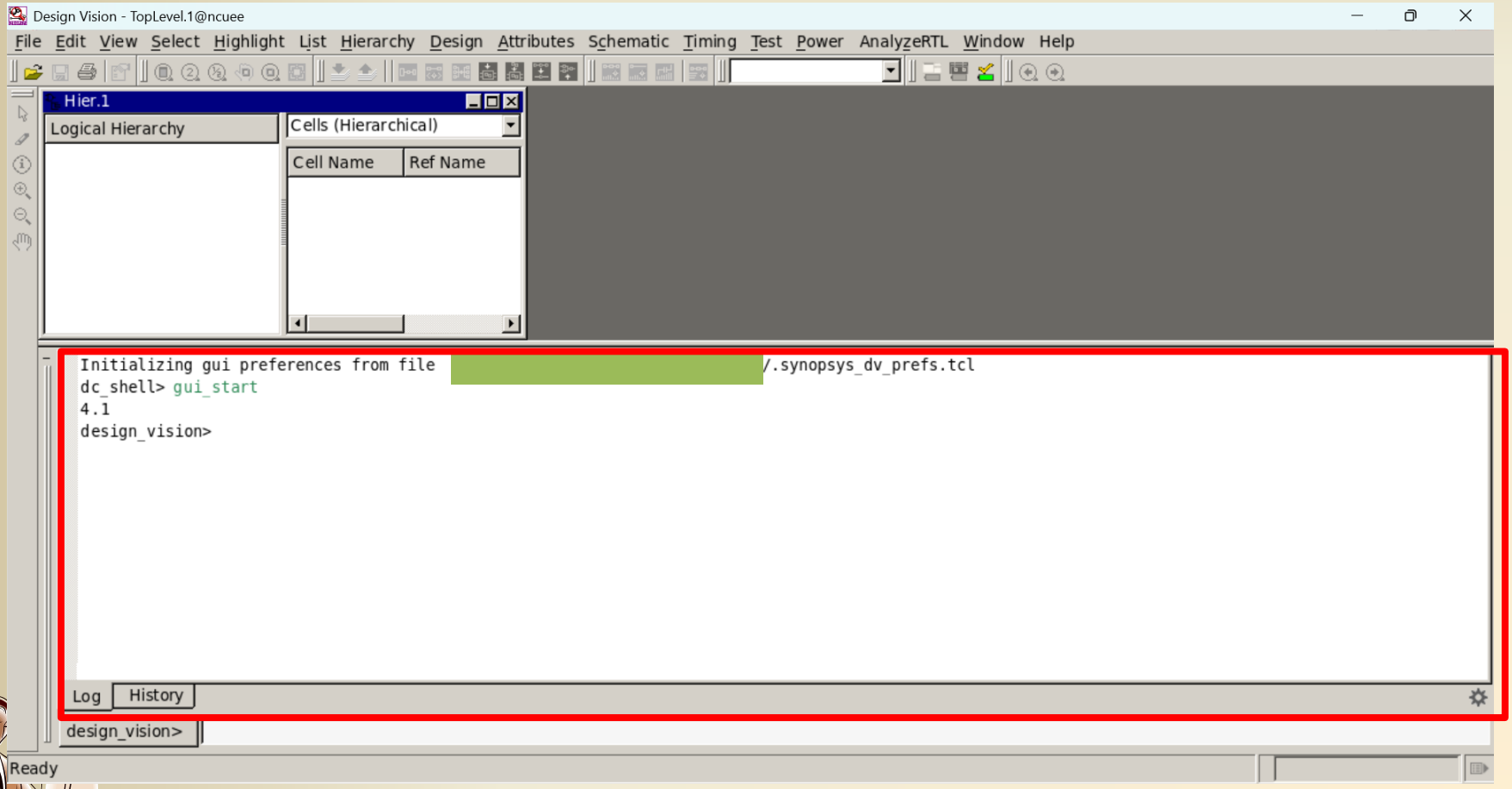
Design Compiler(DC)

DC GUI



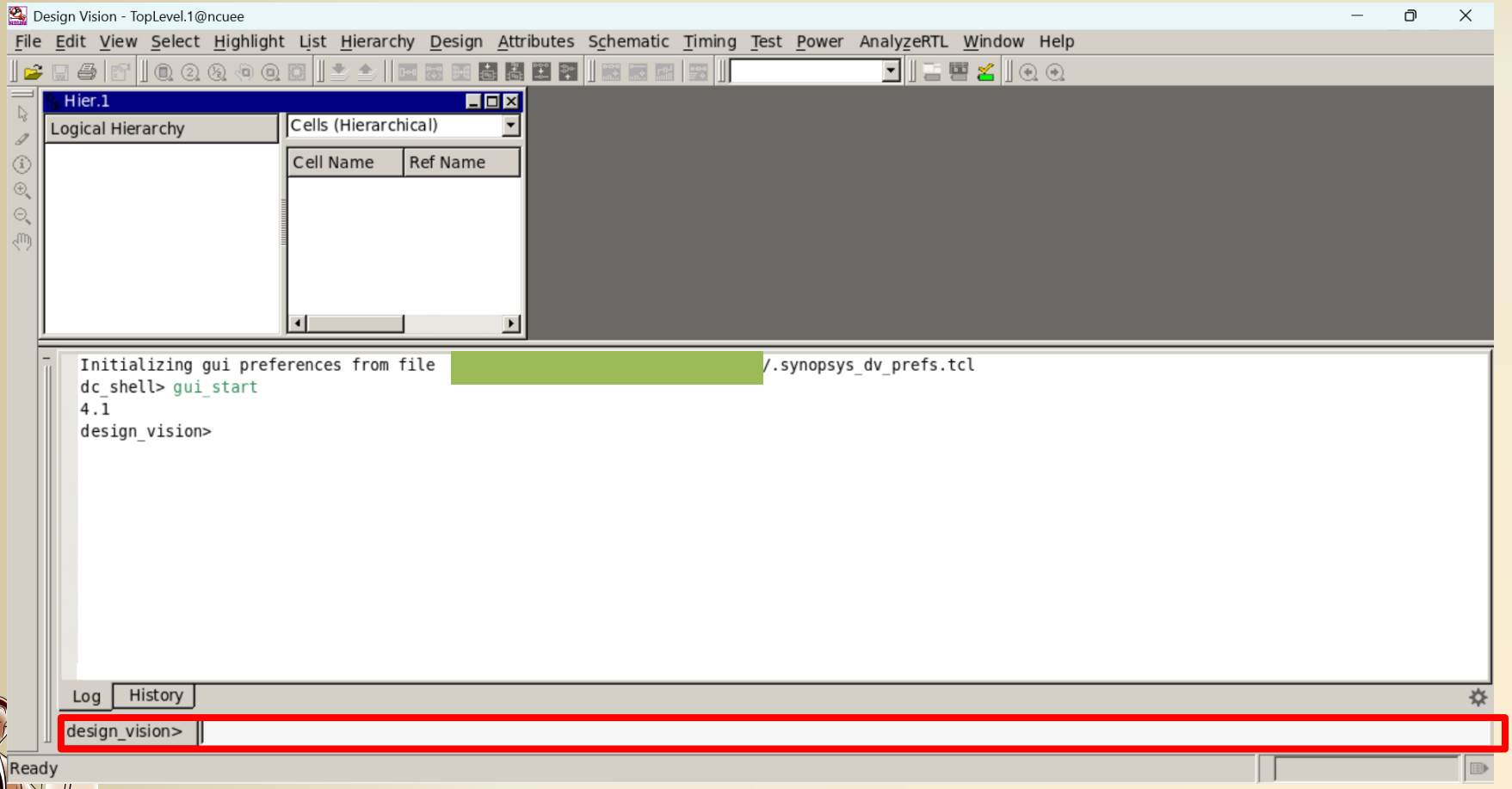
Design Compiler(DC)

Log information



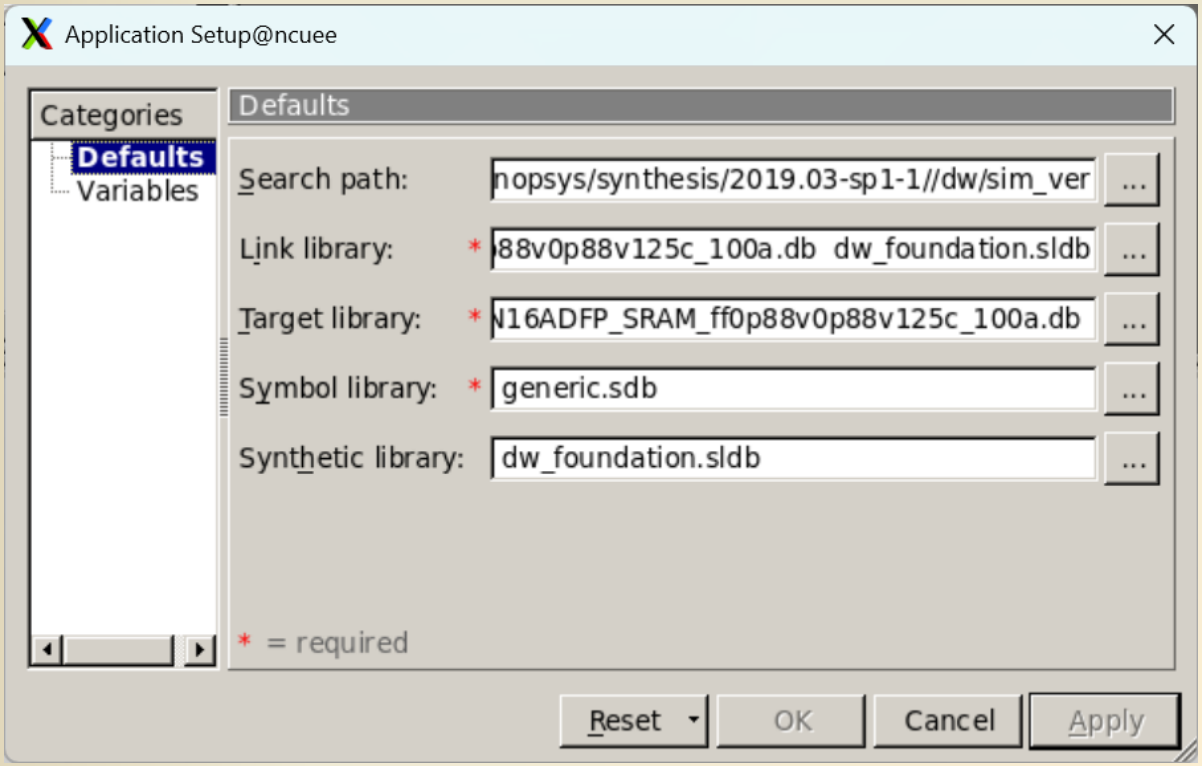
Design Compiler(DC)

Terminal: design vision



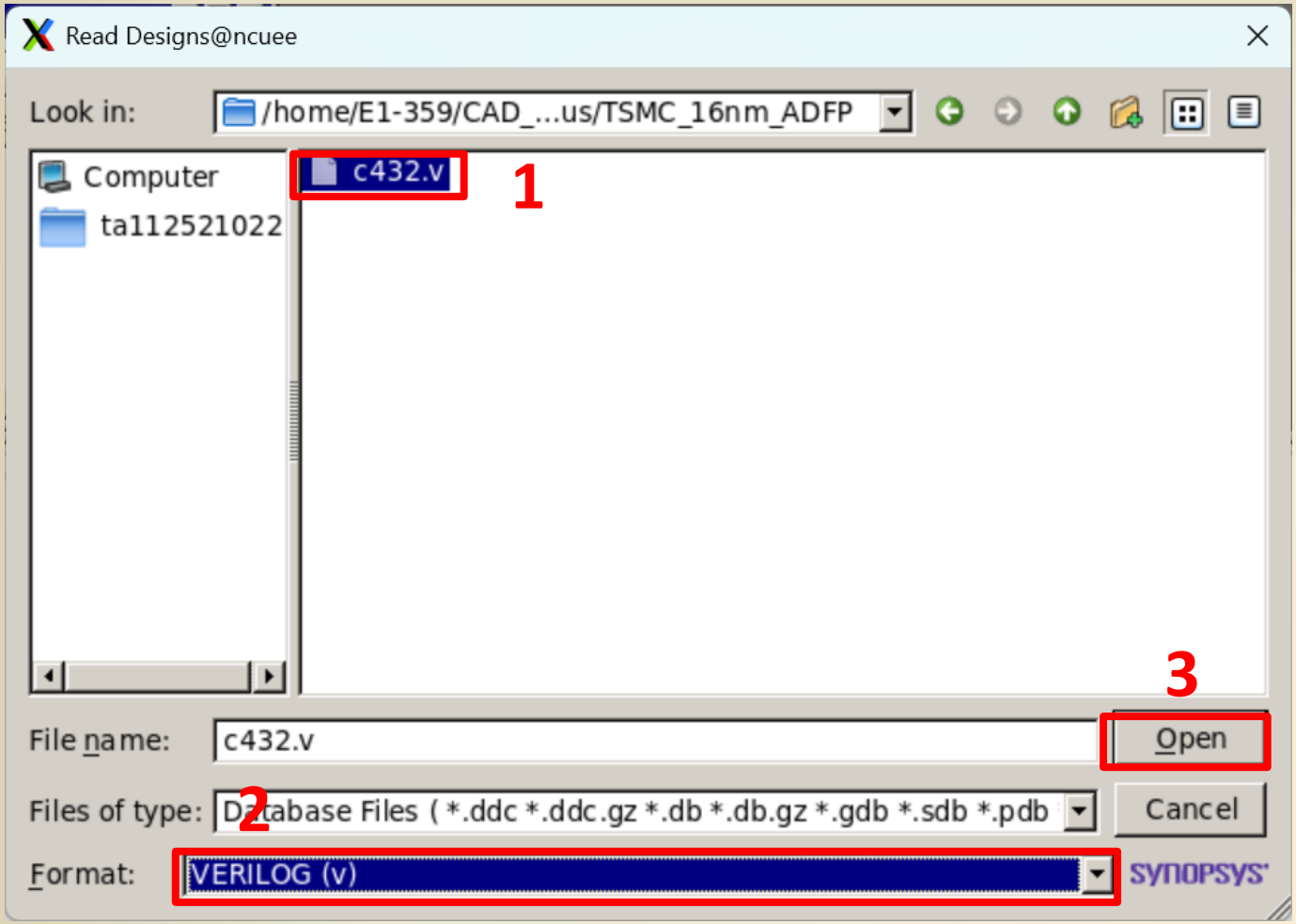
Design Compiler

Step1. Check setup (File > Setup…)



Design Compiler

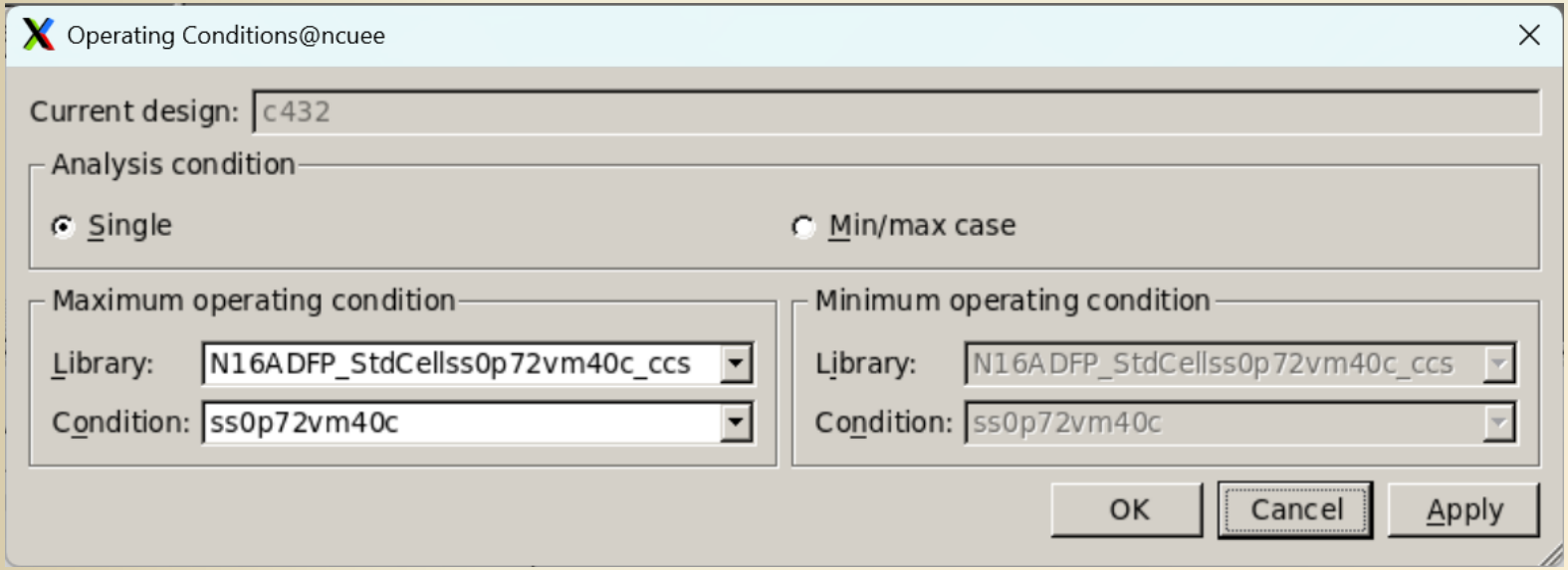
Step2. Read design files (File > Read …)



Design Compiler

Step3. Setting operating conditions

(Attributes > Operating Environment > Operating Conditions…)



Design Compiler

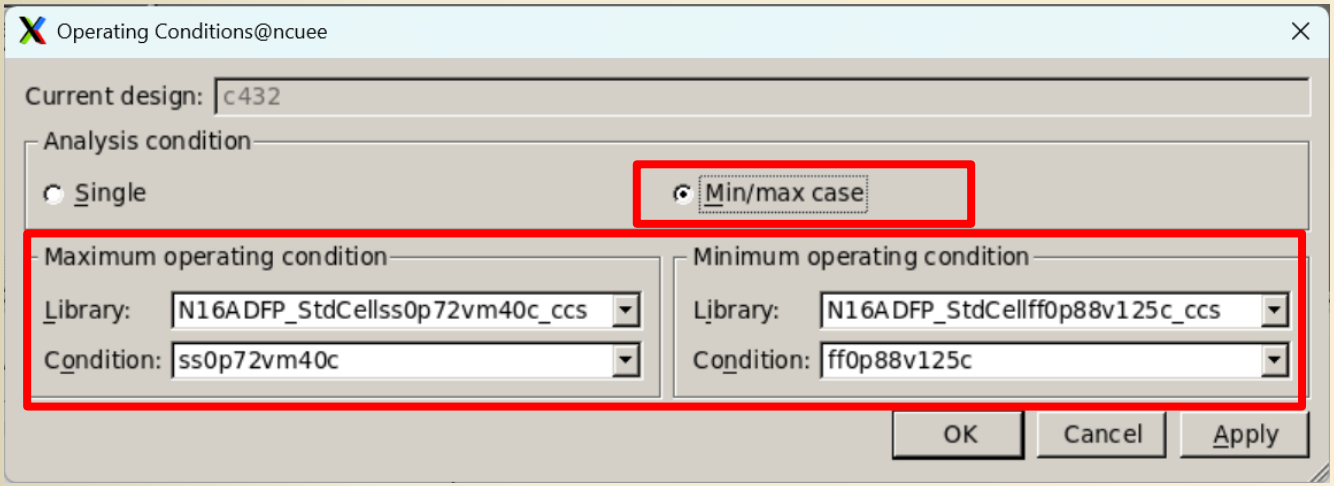
Setting operating conditions

1. **Maximum operating condition** (for setup time analysis):

TSMC_16nm_ADFP	Library: N16ADFP_StdCellss0p72vm40c_ccs, Condition: ss0p72vm40c
TSMC_130nm	Library: slow, Condition: slow

2. **Minimum operating condition** (for hold time analysis):

TSMC_16nm_ADFP	Library: N16ADFP_StdCellff0p88v125c_ccs, Condition: ff0p88v125c
TSMC_130nm	Library: fast, Condition: fast

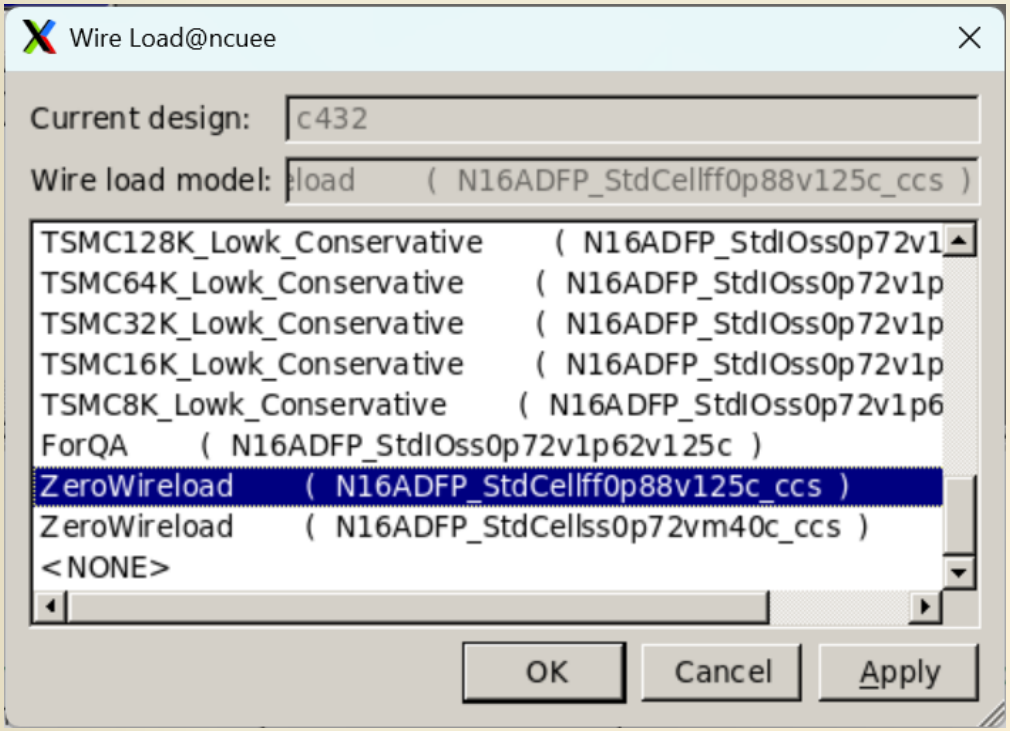


Design Compiler

Step4. Choose ZeroWireload (N16ADFP_StdCellss0p72vm40c_ccs)

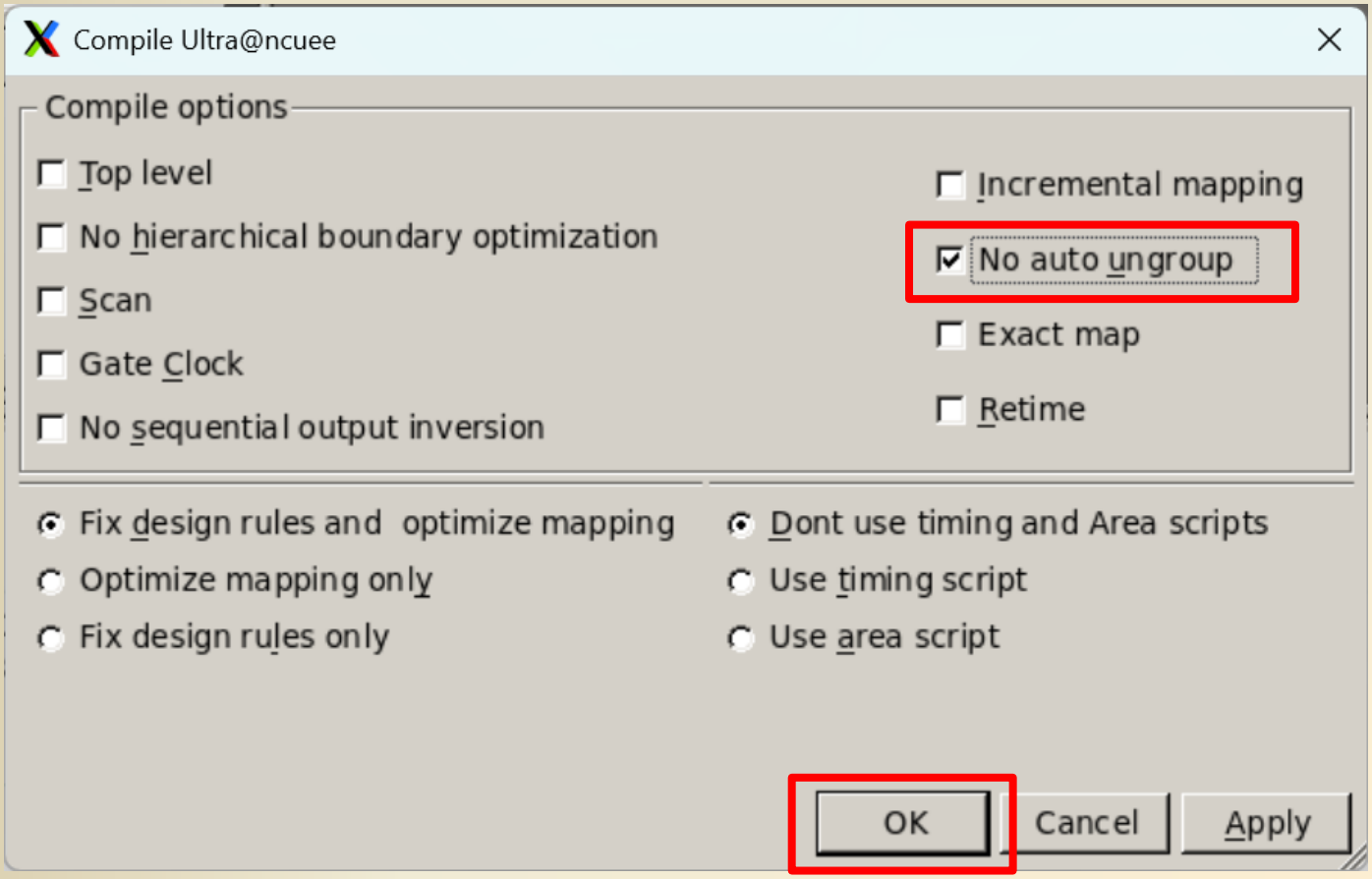
(Attributes > Operating Environment > Wire Load)

(**ADFP needs to do this step. Otherwise, skip it.**)



Design Compiler

Step5. Design > Compile Ultra



Design Compiler

Step6. Report the area. Type the following command.

```
design_vision> report_area > FileName
```

Technology	Name of the report file (FileName)
TSMC_16nm_ADFP	c432_16nm.area
TSMC_130nm	c432_130nm.area

Q1. Total cell area = ____ (one for 130nm, another for 16nm)

Q2. Compare two values. Which one is smaller?

Design Compiler

In the area report file,

```
Number of ports:                          43
Number of nets:                          121
Number of cells:                          85
Number of combinational cells:            85
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                        21
Number of references:                     19

Combinational area:
Buf/Inv area:
Noncombinational area:
Macro/Black Box area:
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:
Total area:                 undefined
```

Have a screenshot like above. Put this in your report. (one for 130nm, another for 16nm)

Design Compiler

Step7. Report the power. Type the following command.

```
design_vision> report_power > FileName
```

Technology	Name of the report file (FileName)
TSMC_16nm_ADFP	c432_16nm.power
TSMC_130nm	c432_130nm.power

Q3. Total power = ____ (one for 130nm, another for 16nm)

Q4. Compare two values. Which one is smaller?

Design Compiler

In the power report file,

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.0000	0.0000	0.0000	0.0000	(0.00%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	[redacted]	[redacted]	[redacted]	[redacted]	(100.00%)	
Total	[redacted] mW	[redacted] mW	[redacted] nW	[redacted] mW		

Have a screenshot like above. Put this in your report.

(one for 130nm, another for 16nm)

Design Compiler

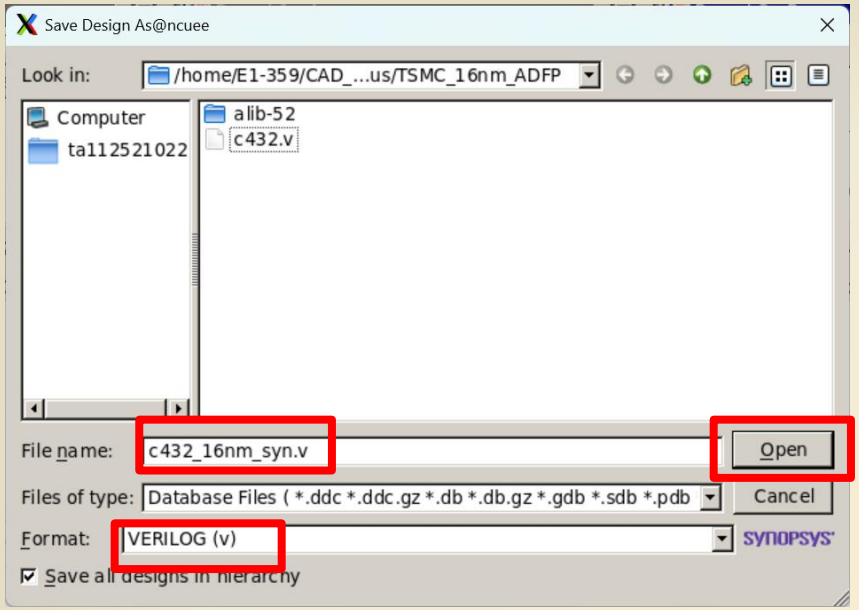
Step8. Generate Legal Verilog Netlist

(In command line)

design_vision> change_names -hierarchy -rules verilog

Design Compiler

Step9. Save Gate-Level Netlist (File > Save as)



Technology	Name of the Verilog after synthesis
TSMC_16nm_ADFP	c432_16nm_syn.v
TSMC_130nm	c432_130nm_syn.v

Design Compiler

Step10. Write Standard Delay File (SDF)

(In command line)

design_vision> write_sdf -version 2.1 -load_delay cell FileName

Technology	Name of the SDF (FileName)
TSMC_16nm_ADFP	c432_16nm_syn.sdf
TSMC_130nm	c432_130nm_syn.sdf

Design Compiler

Step11. Close Design Vision GUI

(In command line)

design_vision> quit

Design Compiler

Step12. Edit testbench (c432_testbench.v) → Insert commands

`define SDFFILE "SDF Name"

```
1   `timescale 1ns / 1ps
2   `define SDFFILE "c432_16nm_syn.sdf"
3   ////////////////////////////////////
4   ////////////////////////////////////
```

`ifdef SDF

initial $sdf_annotate(`SDFFILE, uut);

`endif

```
16  c432 uut(.N1(N1),.N4(N4),.N8(N8),.N11(N11),.N14(N14),.N17(N17),.N21(N21),.N24(N24),.N27  ),.N30(N30),.N34(N34),
17
18  `ifdef SDF
19      initial $sdf_annotate(`SDFFILE, uut);
20  `endif
```

Design Compiler

Step13. Verify the synthesis result

```
$ source /usr/cad/synopsys/CIC/verdi.cshrc
$ source /usr/cad/cadence/CIC/incisiv.cshrc
$ ncverilog testbench.v design_syn.v simulation_model.v +define+SDF
```

For TSMC_16nm_ADFP,

Item	Name/Directory
testbench.v	c432_testbench.v
design_syn.v	c432_16nm_syn.v
simulation_model.v	"/home/adfp_tools/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v"

Design Compiler

Step13. Verify the synthesis result

```
$ source /usr/cad/synopsys/CIC/verdi.cshrc
$ source /usr/cad/cadence/CIC/incisiv.cshrc
$ ncverilog testbench.v design_syn.v simulation_model.v +define+SDF
```

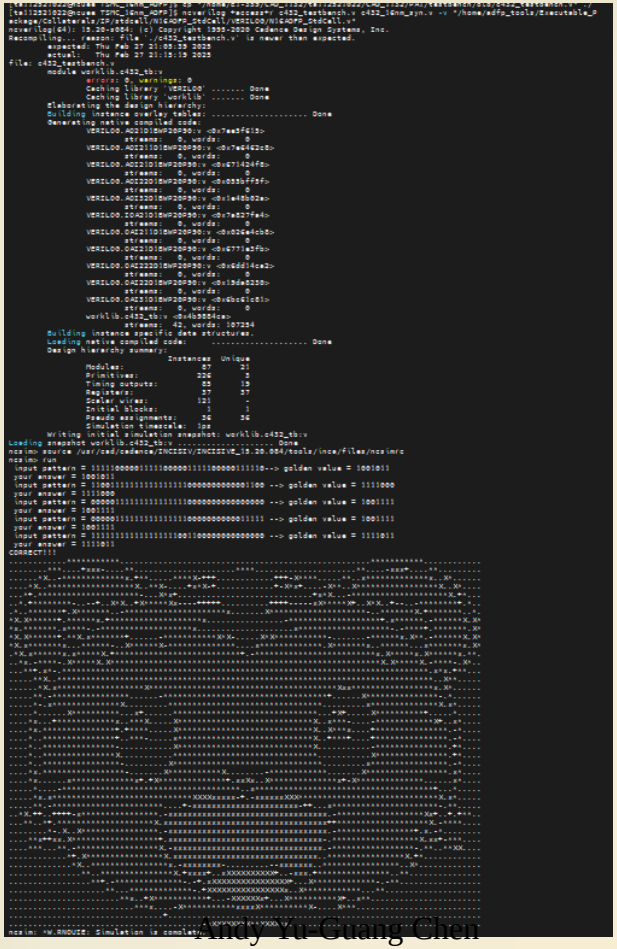
For TSMC_130nm,

Item	Name/Directory
testbench.v	c432_testbench.v
design_syn.v	c432_130nm_syn.v
simulation_model.v	"/home/tools/designkit/CBDK_IC_Contest_v2.1/Verilog/tsmc13.v"

Design Compiler

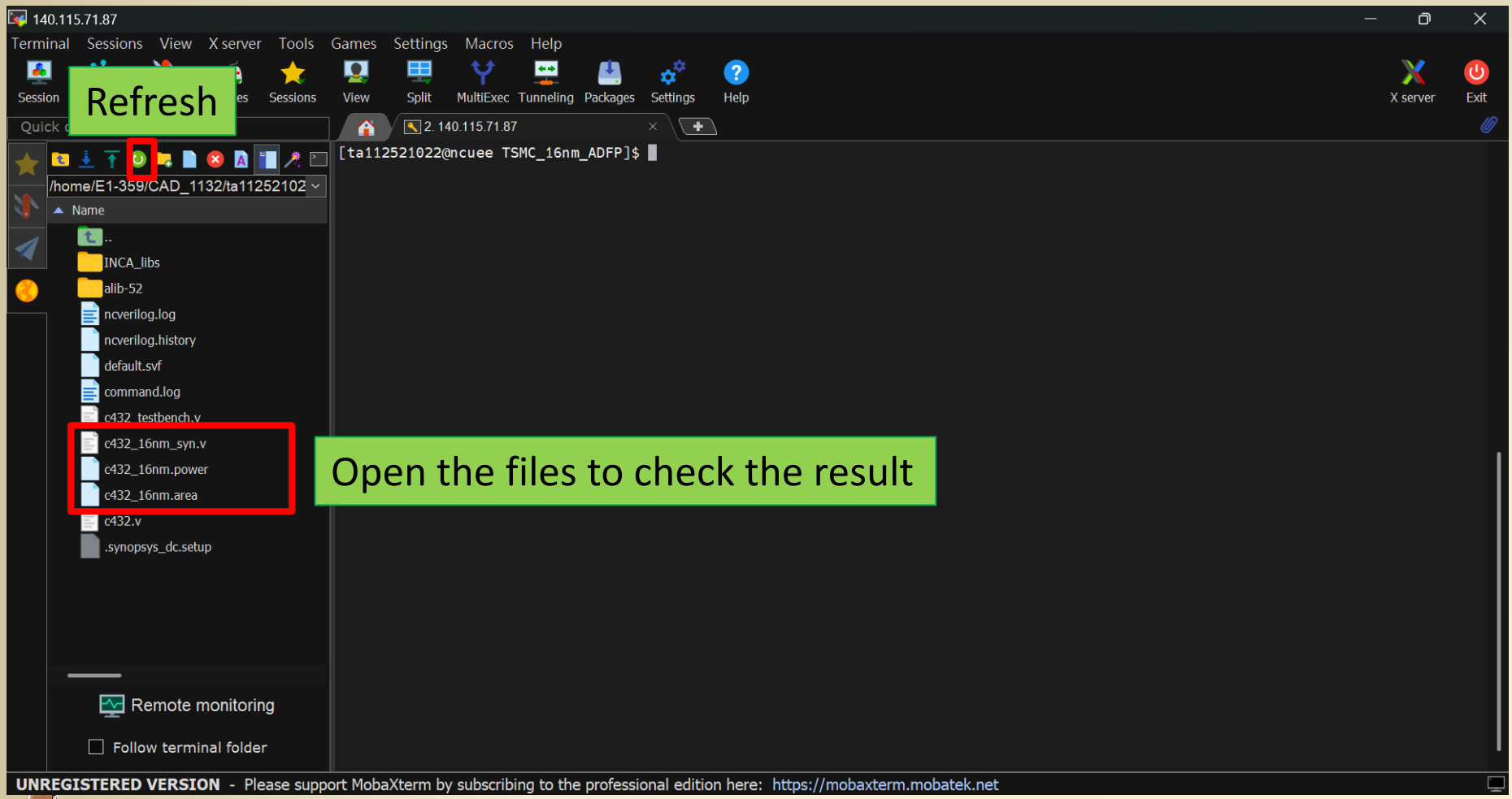
Q5. Check whether the synthesis result is correct.

Have a screenshot of the result. (one for 130nm, another for 16nm)



Where to find the report files ?

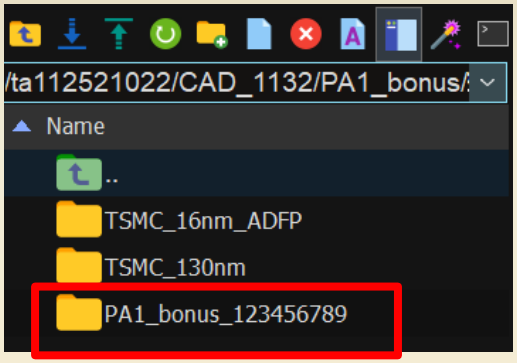
Report file


140.115.71.87
Terminal Sessions View X server Tools Games Settings Macros Help
Session
Refresh
Sessions
View
Split
MultiExec
Tunneling
Packages
Settings
Help
X server
Exit
2. 140.115.71.87
[ta112521022@ncuee TSMC_16nm_ADFP]$
/home/E1-359/CAD_1132/ta11252102
Name
..
INCA_libs
alib-52
ncverilog.log
ncverilog.history
default.svf
command.log
c432_testbench.v
c432_16nm_syn.v
c432_16nm.power
c432_16nm.area
c432.v
.synopsys_dc.setup
Open the files to check the result
Remote monitoring
Follow terminal folder
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

How to compress the files?

Step 1: Create a folder, named as PA1_bonus_StudentID

$ mkdir ../PA1_bonus_123456789



Step 2: Copy all the required files into this folder

$ cp c432_16nm.area ../PA1_bonus_123456789

$ cp c432_16nm.power ../PA1_bonus_123456789

$ cp c432_16nm_syn.v ../PA1_bonus_123456789

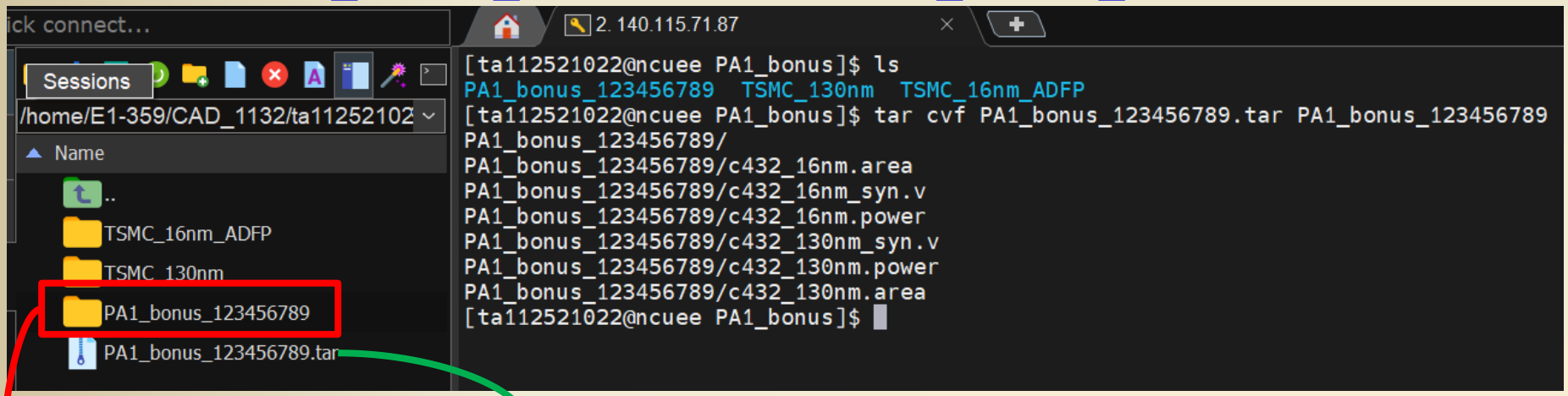
$ cp c432_16nm_syn.sdf ../PA1_bonus_123456789

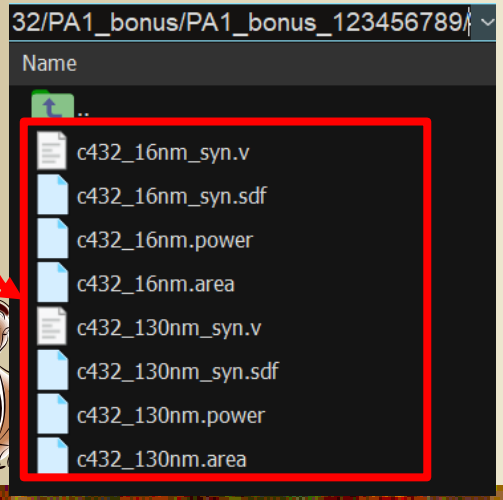
⋮

How to compress the files?

Step 3: Compress the folder into .tar file

$ tar cvf PA1_bonus_123456789.tar PA1_bonus_123456789





You have to submit this .tar file